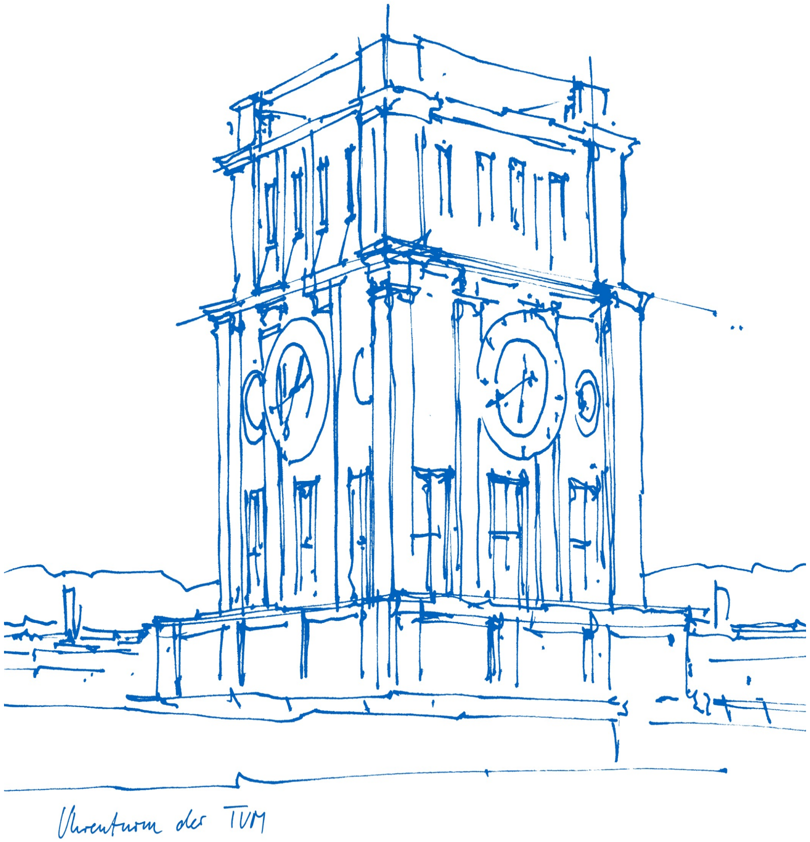
Geoinformation

Prof. Dr.-Ing. Liqiu Meng

Chair of Cartography and Visual Analytics

WS 25/26



Course coordinator

Mr. Zihan Liu khan.liu@tum.de

Lecture blocks

L1-2: Introduction

L3-4: Spatiotemporal representations and databases

L5-7: Spatial data analysis

L8-9: Selected cartographic techniques

L1: Introduction – Part 1

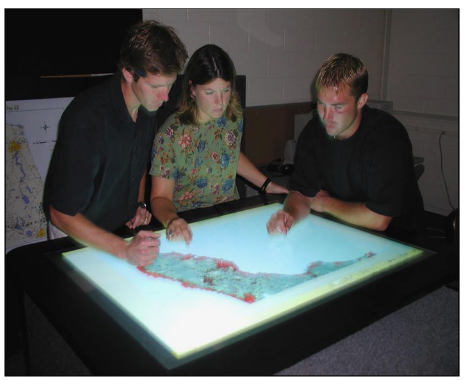
I. An overview of geoinformatics

II. The nature of geoinformation

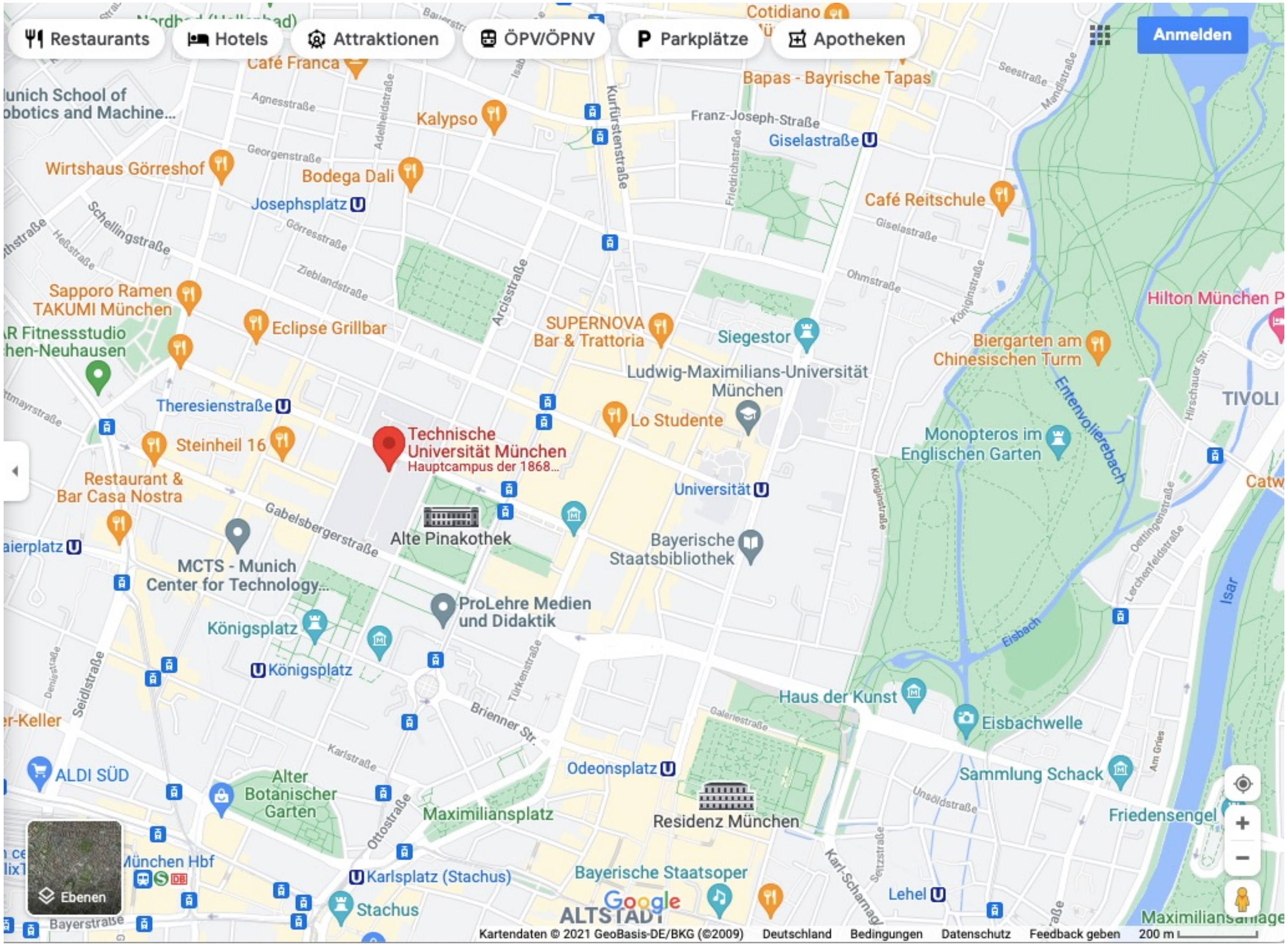
I. An overview of geoinformatics

Geoinformation in our daily life

Citizens and professionals typically want to know and reason about location, time, object, process etc.

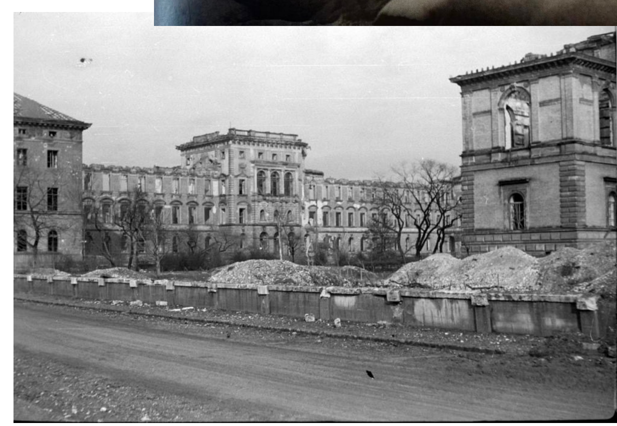


Geoinformation about individual PoIs

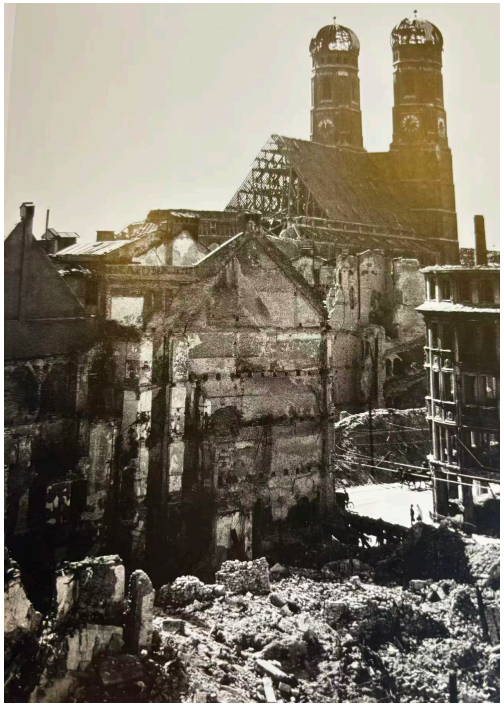


TUM 1874

TUM 1945



Alte Pinakothek 1836

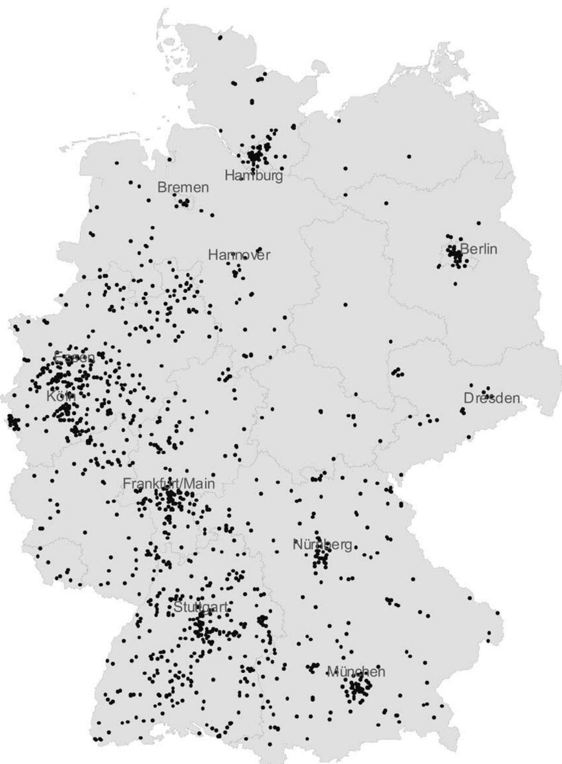



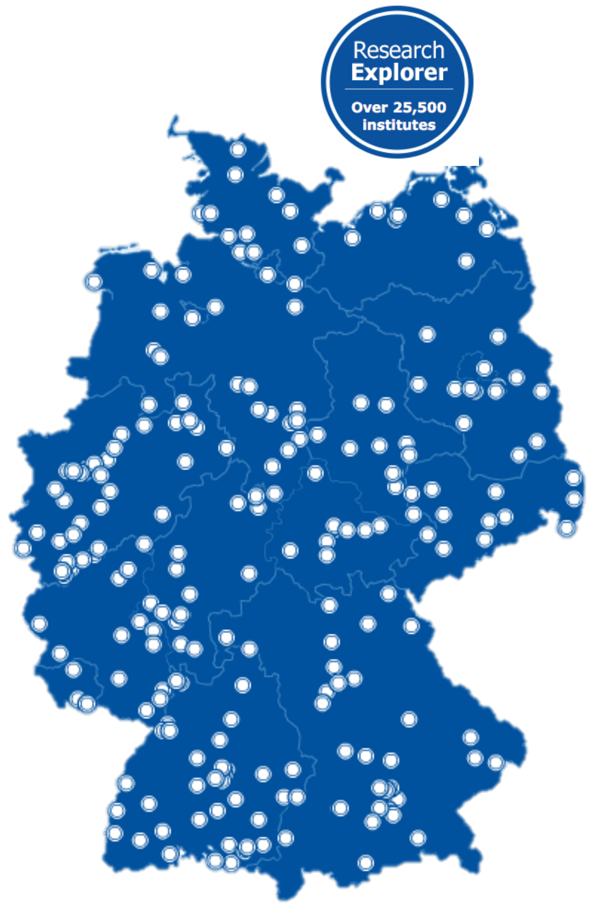



Olympia hill in Munich

Geoinformation about distributions

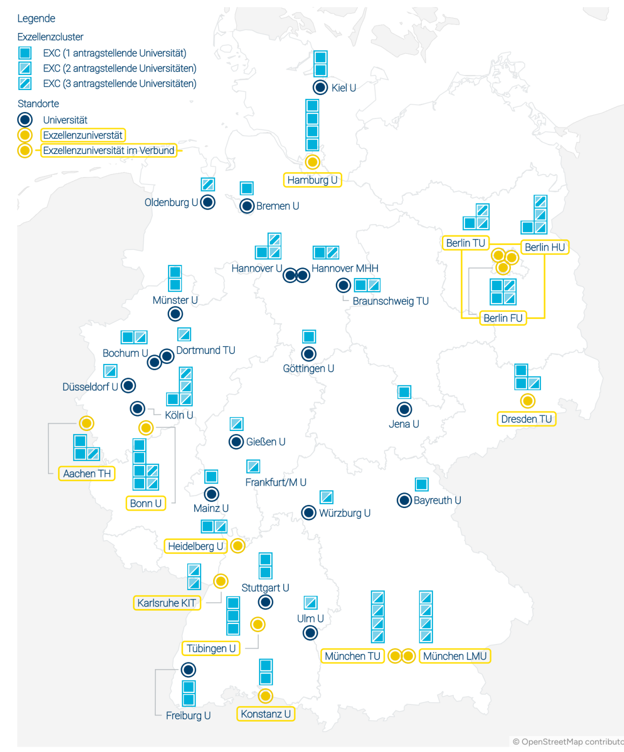
Hidden champions





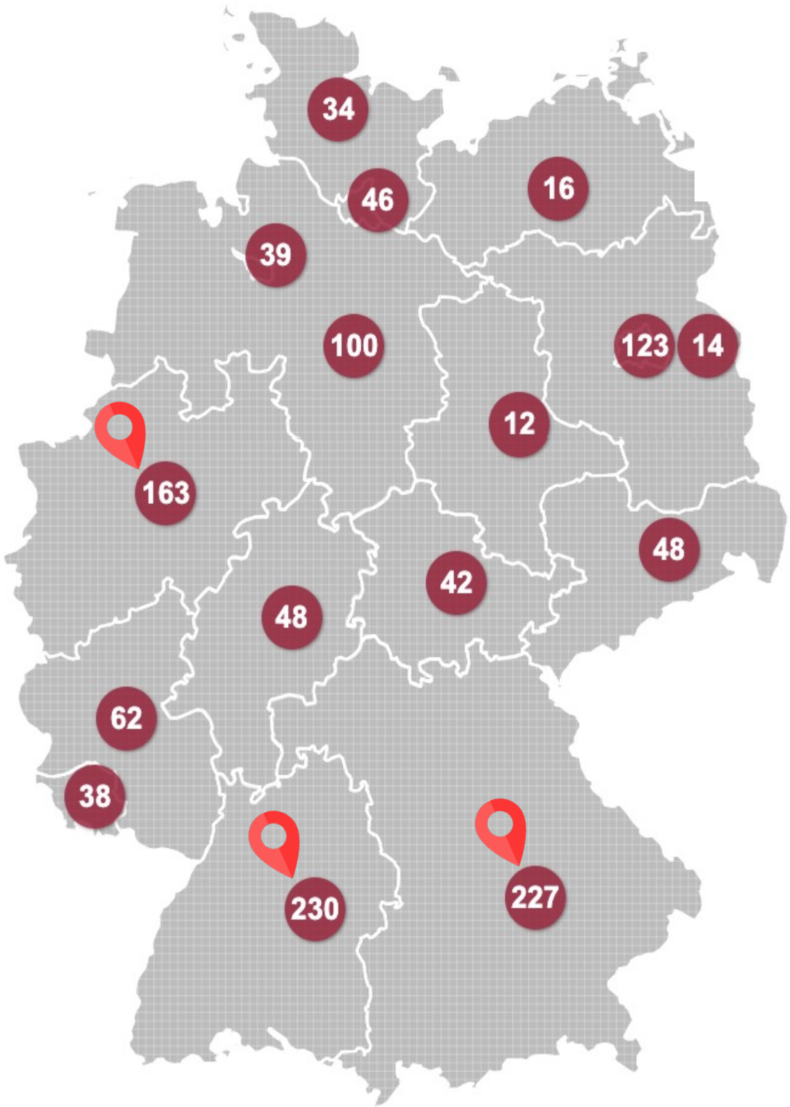
Handelsblatt

Elite universities

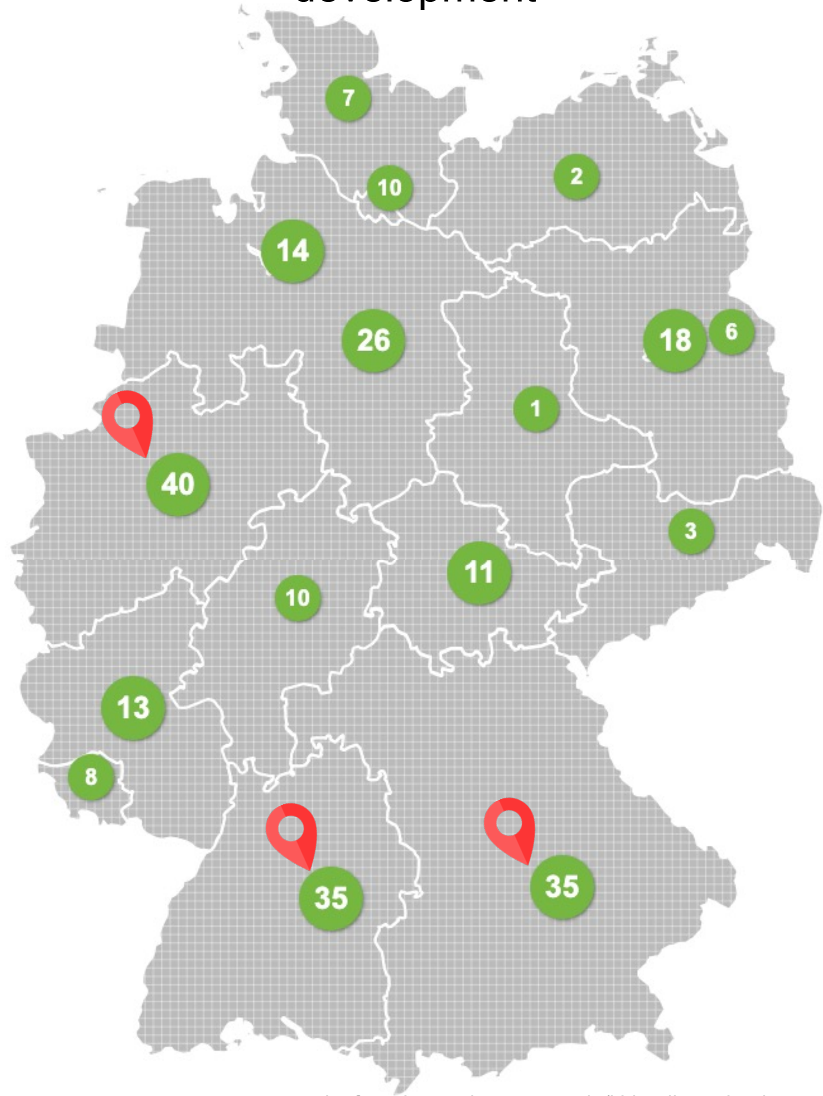


Grafik: Deutsche Forschungsgemeinschaft (DFG). Karte der Bundesländer: © GeoBasis-DE / BKG 2024 (verändert).

Best practices AI projects



AI projects for sustainable development



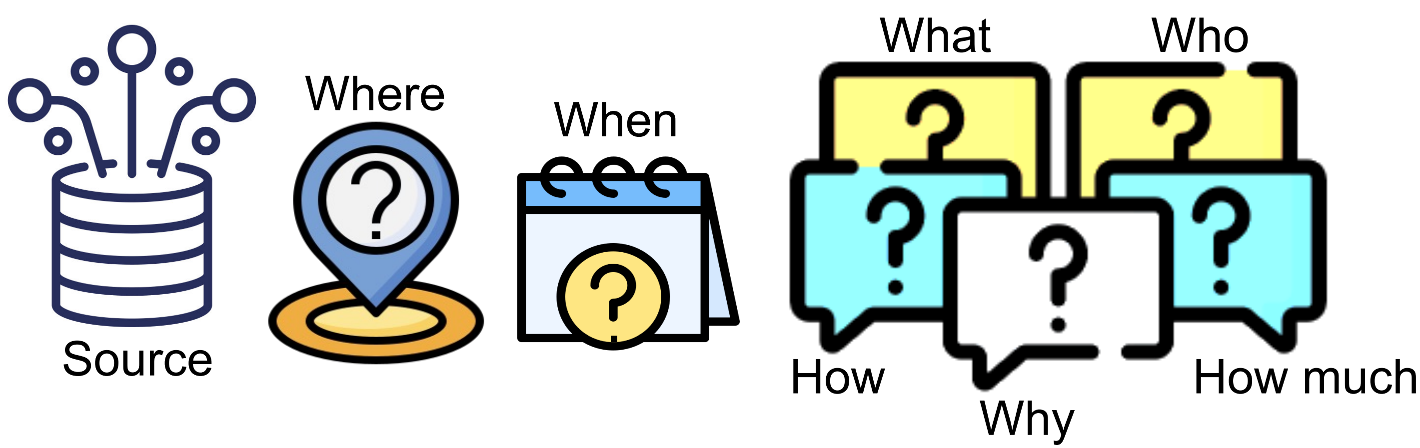
www.plattform-lernende-systeme.de/ki-landkarte.html

Geoinformation about events

Fictive news 1 from Daily ABC, Oct 18, 2025

A recent analysis of satellite imagery has confirmed what local farmers have claimed for months: the White Nile River has shifted its course. Over the past two years, a combination of severe drought and a massive landslide 500 km upstream has caused the river to carve a new channel, moving ca. 2 km to the east.

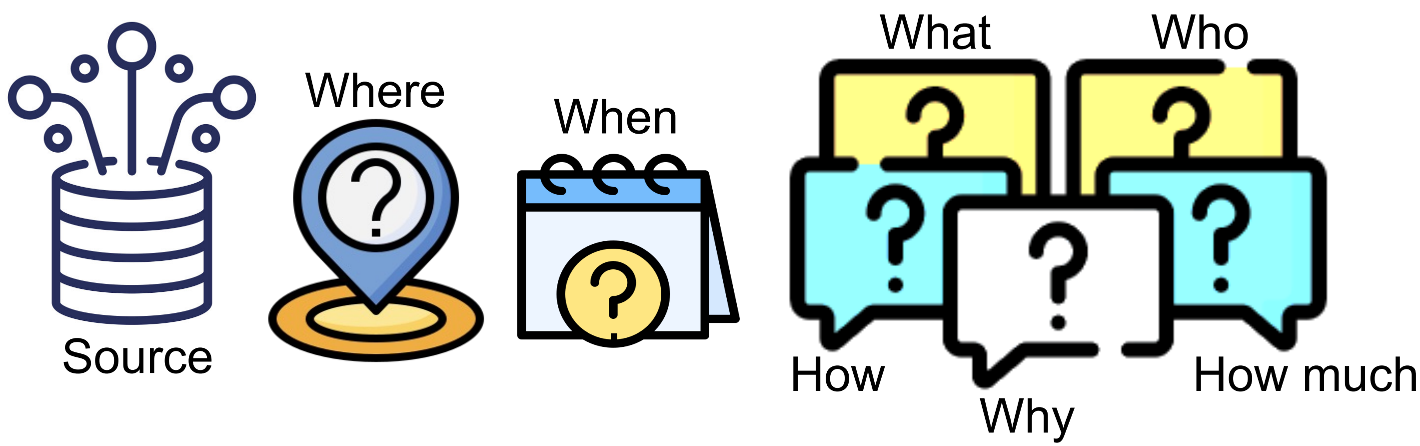
The new river course has placed a 50 km^2 area of fertile farmland, previously part of South Sudan, on the eastern bank, which is sovereign territory of Ethiopia. The farmers who have worked this land for generations now find themselves in a different country without moving an inch. The foreign ministries of both nations have to resolve the situation, highlighting how dynamic processes can challenge static political borders.



Fictive news 2 from Daily ABC, Oct 19, 2025

The city council of Phoenix has passed a "Cool Roofs" mandate following the release of a startling heat map of the city. The map, created by researchers at Arizona State University, used satellite data to reveal a temperature difference of up to 8°C between the dense, paved city center and the surrounding desert and parklands — a typical Urban Heat Island effect. “It‘s not enough to know where things are, we have to now *why* they are that way," said Dr. Lena Reyes, the lead researcher. “

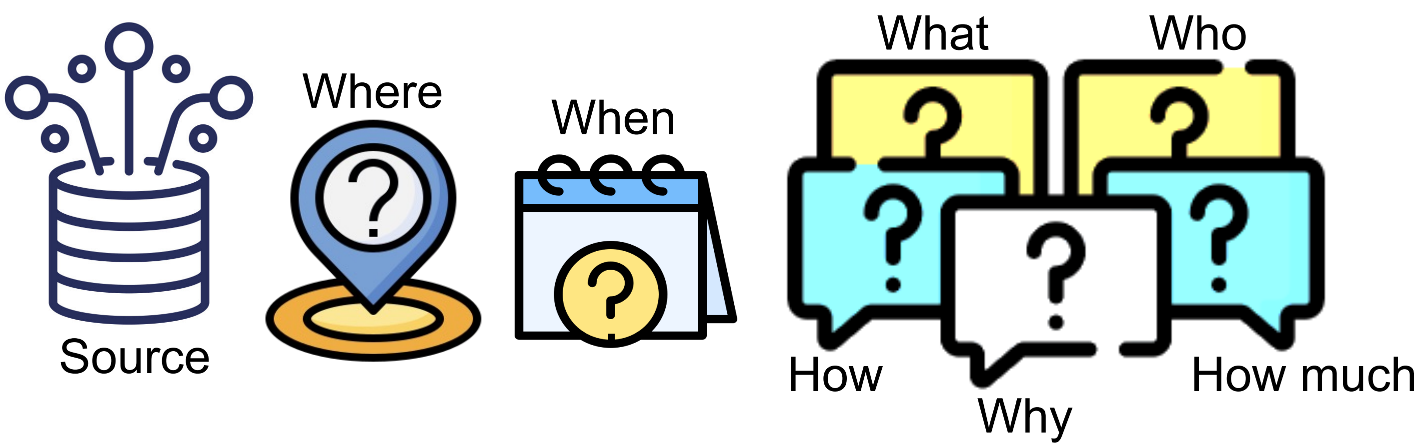
Under the new policy, all new commercial buildings and major renovations within the city's urban core must be constructed with light-colored, reflective roofing materials. The city is also offering subsidies to homeowners in these areas to repaint their roofs.



Fictive news 3 from Daily ABC, Oct 20, 2025

A massive construction project for a new residential complex in central Berlin was abruptly halted today after workers unearthed a 500-kg unexploded bomb from World War II. The discovery occurred at a site on the corner of Wilhelmstrasse and Niederkirchnerstrasse. The bomb was located using a combination of historical aerial photographs from 1945 and modern ground-penetrating radar. Experts from the German bomb disposal unit, have cordoned off a 1,000 m evacuation zone, affecting over 5,000 residents, and a segment of the U-Bahn subway line.

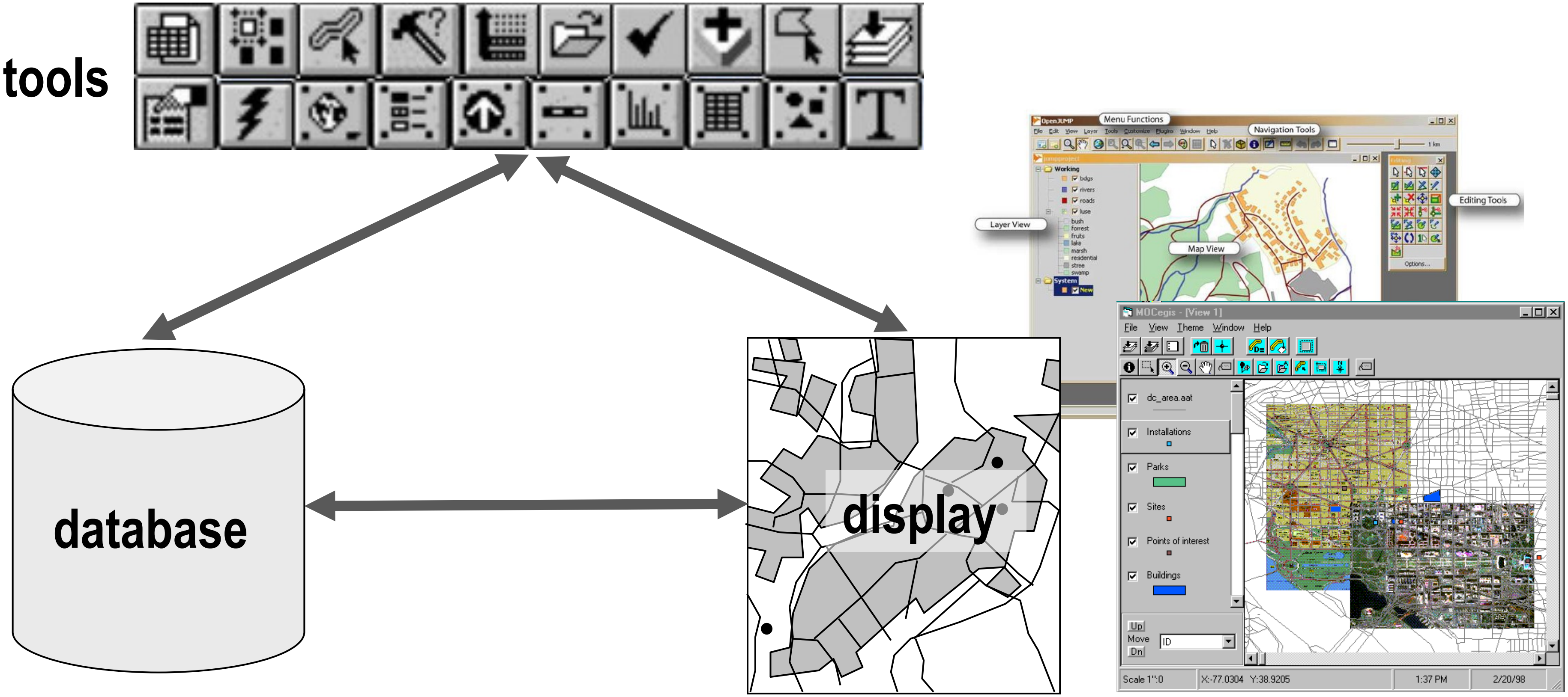
A spokesperson of the German bomb disposal unit stated, "This is a reminder that the past is buried just beneath our feet. Our work relies on overlaying historical maps with modern city plans to keep people safe." Disposal experts expect the defusing operation to take most of the day.



Geoinformatics

A discipline dealing with acquisition, storage, processing, presentation and dissemination of geoinformation incl. the infrastructure to secure efficient and ethical use of this information.

Essential components of a GIS



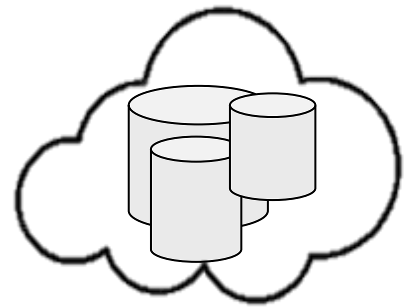
The typical workflow of a GIS

Spatial data modeling

acquisition

integration

updating

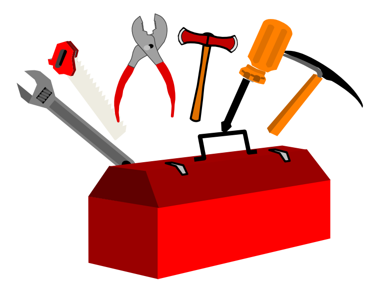


quality assessment

Spatial data handling

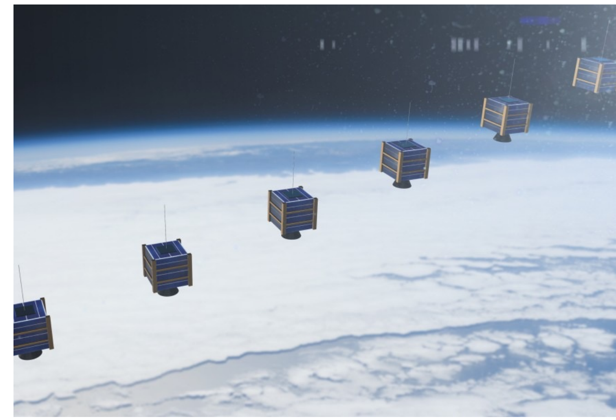
visualization

analysis

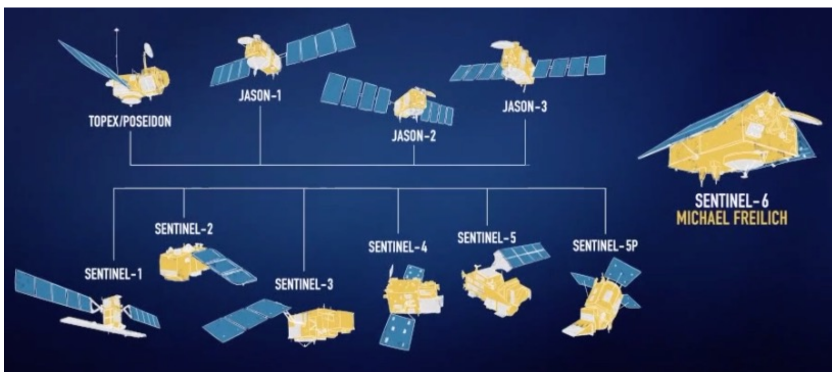


Major sources of geodata

1. Remote sensing with structured Earth observation

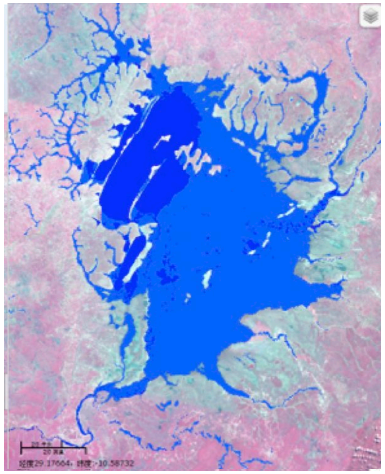


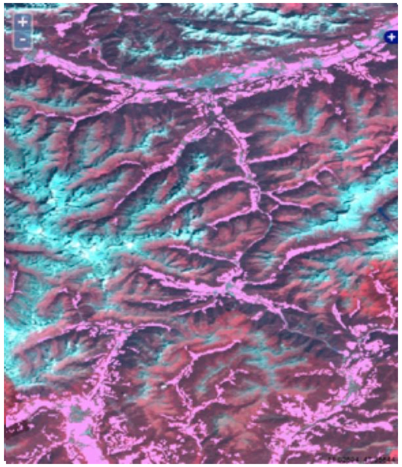
Earth Observation Satellites



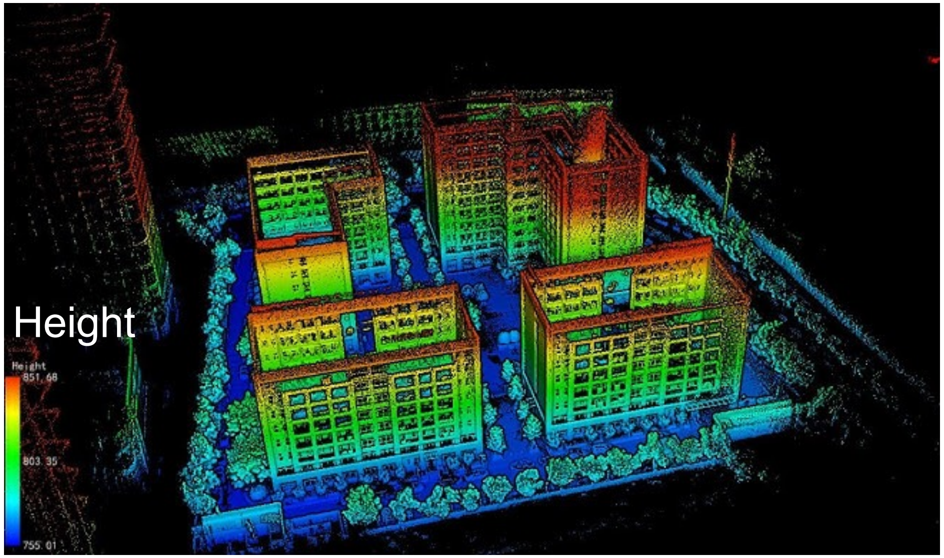
Aerial Photogrammetry

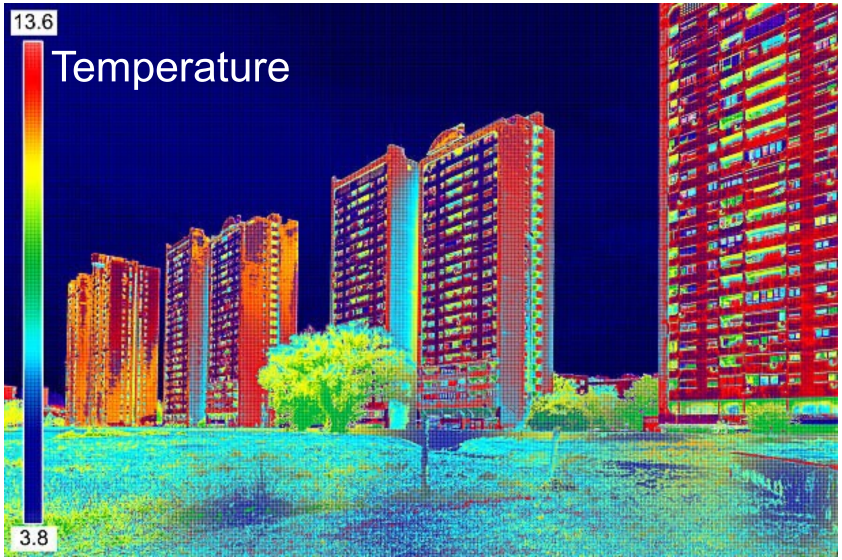
Terrestrial laser scanning





Multispectral images and their interpretation





Learning and deep learning in remote sensing

- Segmentation, classification, recognition
- Change detection
- Quality assurance of geodata
- …

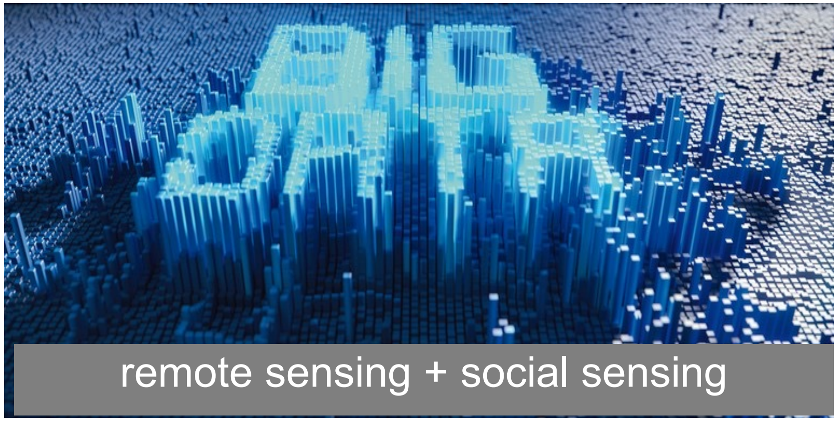
2. Social sensing based on less structured crowd-sourcing

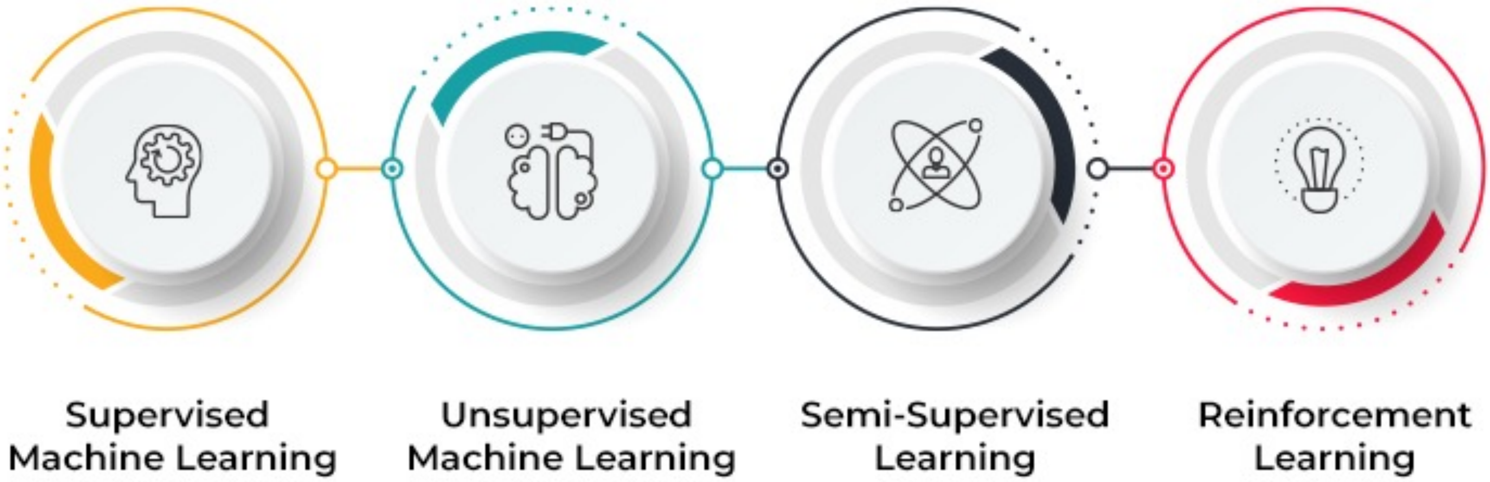
Deep learning in social sensing

Understanding of natural languages, trajectories, and street views

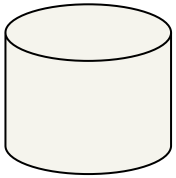
- Natural Language Processing for improved search
- Sentiment analysis for improving user experience and engagement
- Predictive analytics for targeted advertising
- Trajectory analysis and geotargeting for safety and health measures
- Detection of privacy violation
- …

Big geospatial data

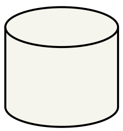




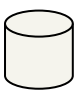
A value cycle from geodata to intelligence



Data - neutral



Information - relevant



Service - useful

Intelligence – beneficial/profitable

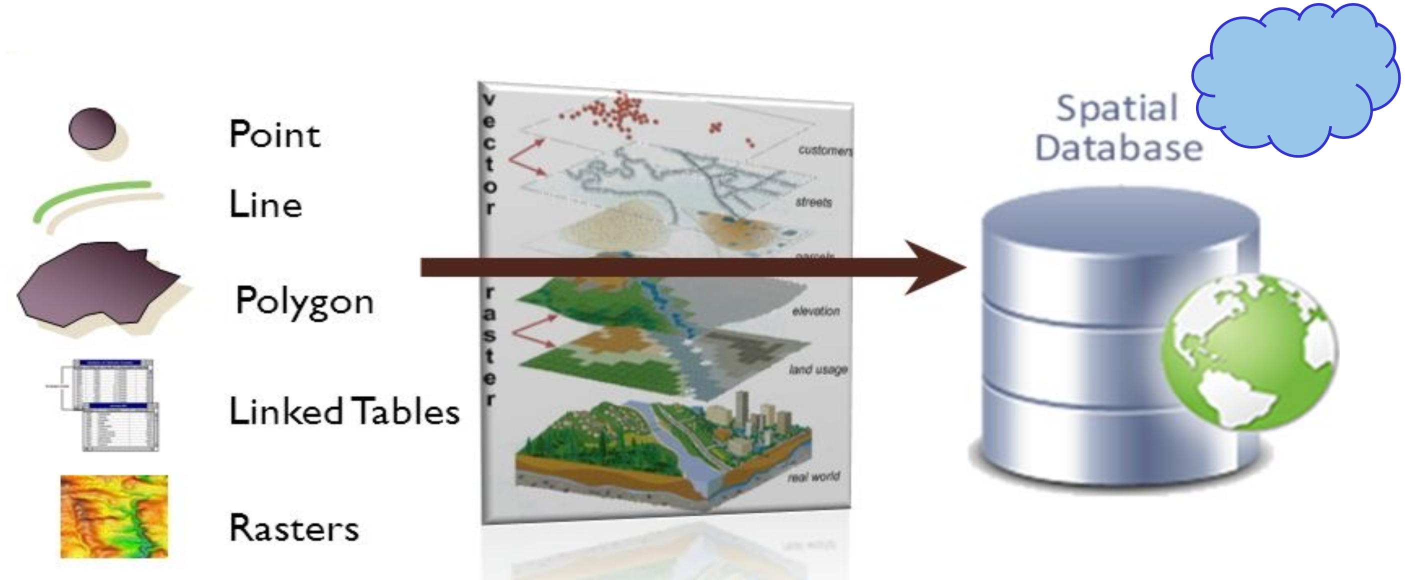
P3 Parkhaus Rathaus 30 Plätze

Einfahrtshöhe:2,00 m (0 Frauenparkplätze)

Öffnungszeit: Mo-Fr 7-19.30 Uhr, Sa 8-17 Uhr

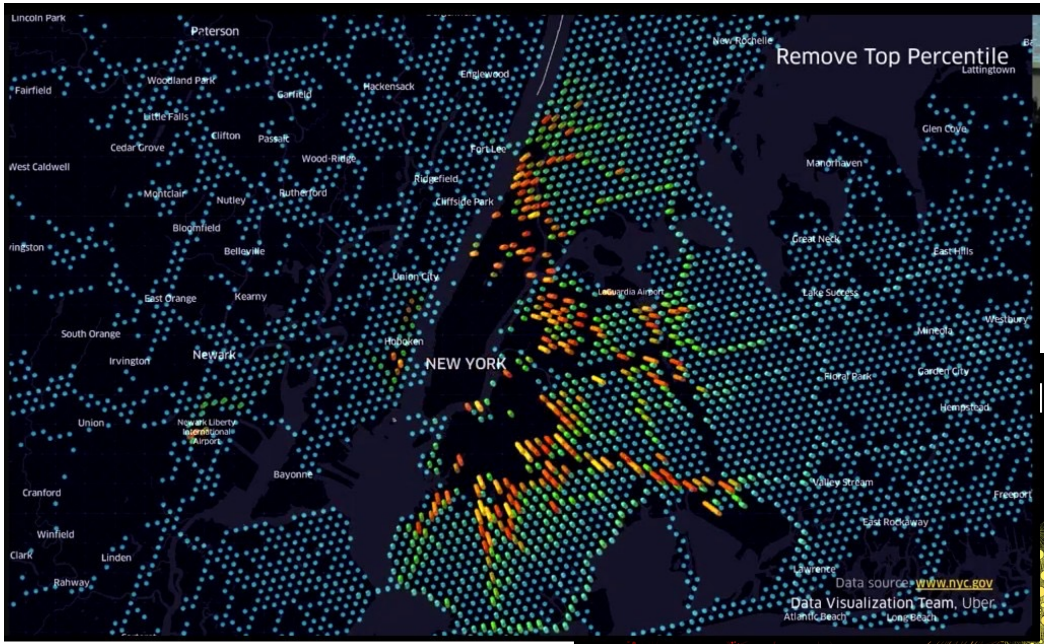
PARKGEBÜHREN	
gesamte Öffnungszeit	
bis 1 Stunde	1,00 €
bis 2 Stunden	2,00 €
bis 3 Stunden	3,00 €
bis 4 Stunden	4,00 €
bis 5 Stunden	4,50 €
ab 5 Stunden	5,50 €
Für Rathausbesucher	
pauschal	0,00 DM
Chips des Cip-Systems werden angenommen (0,50 €)	

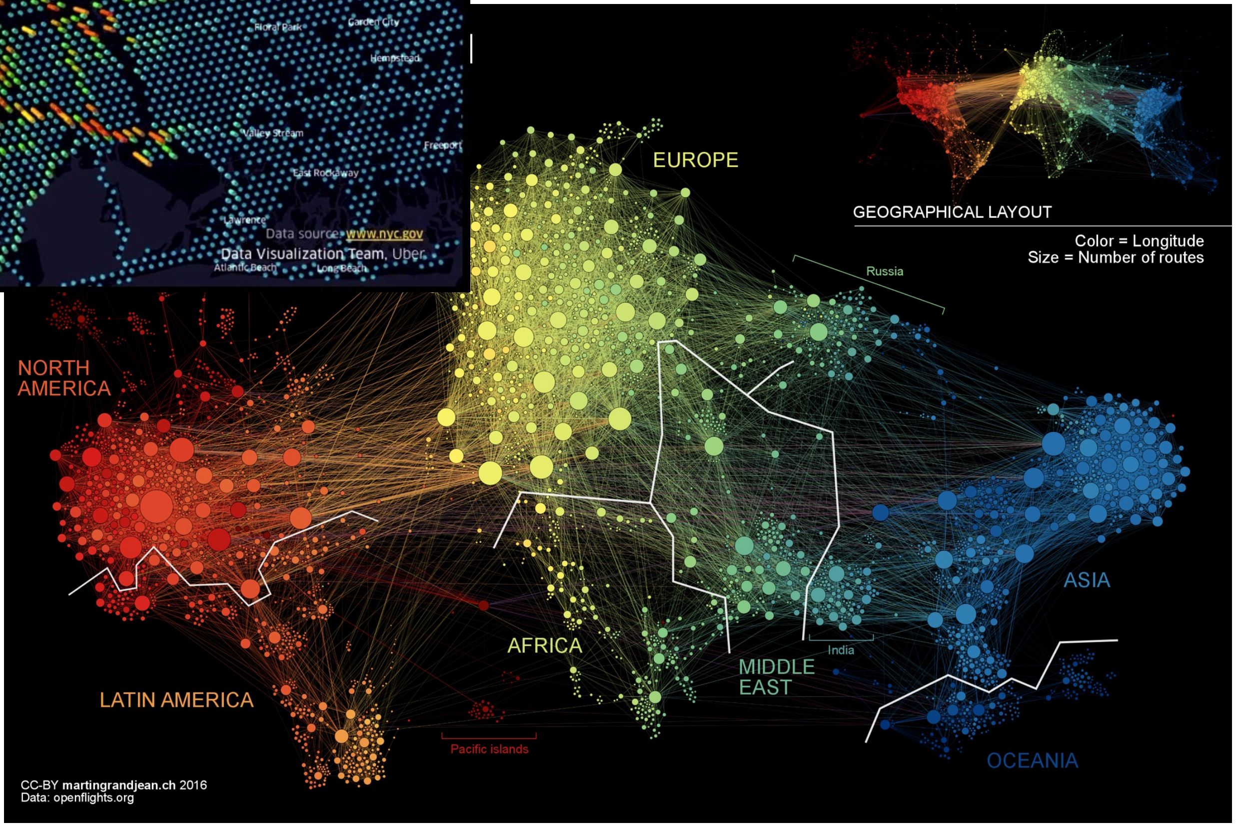
Possibilities of your future career



Geodata manager

Geodata visualizer





domain knowledge
+
machine learning

Geodata scientist / visual storyteller

Milestones in the history of GIS

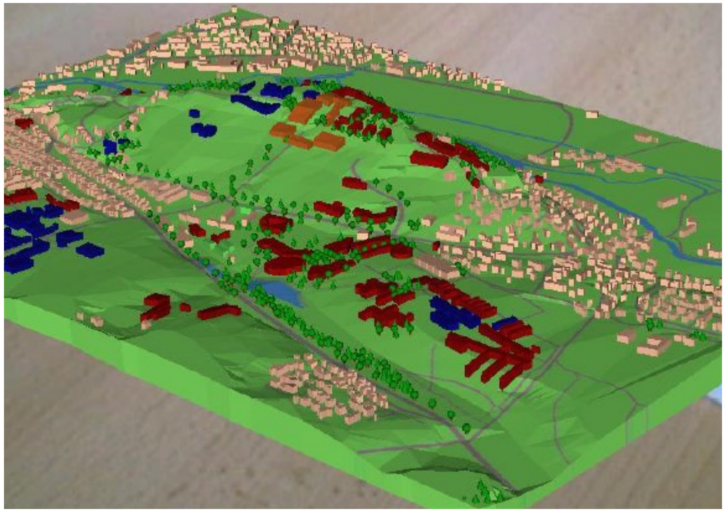
- ❖ Computer-aided analysis and display tools (1950s)
- ❖ Mapping tools (1960s)
- ❖ Emergence of the term “GIS” (1970s)
- ❖ Image processing tools and Graphical User Interfaces (1980s)
- ❖ Distributed GIS and peer-to-peer geo-computing (1990s)
- ❖ Mobile GIS (2000s)
- ❖ Big geodata, cloud computing (2010s)
- ❖ Deep learning, GeoAI (2020s)

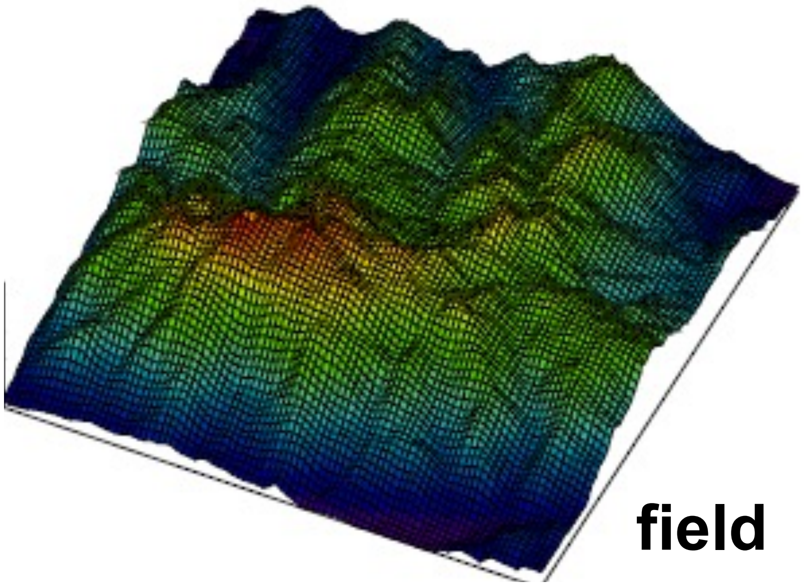
II. The nature of geoinformation

Categories of geoinformation

1. Entities – spatially discrete objects that statically litter in space (e.g. cities, roads)
2. Fields – spatially continuous functions with a unique value everywhere (e.g. terrain height, temperature, moisture)
3. Phenomena – temporally continuous functions that happen dynamically (e.g. brush fires, epidemic, urban growth)

entities





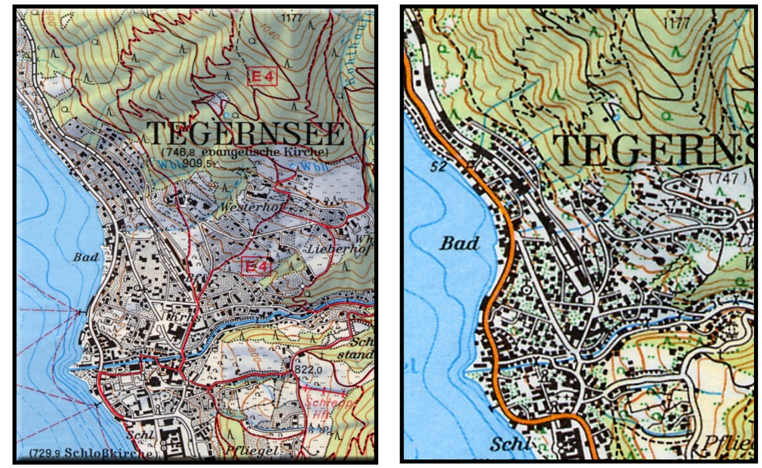
field

phenomena

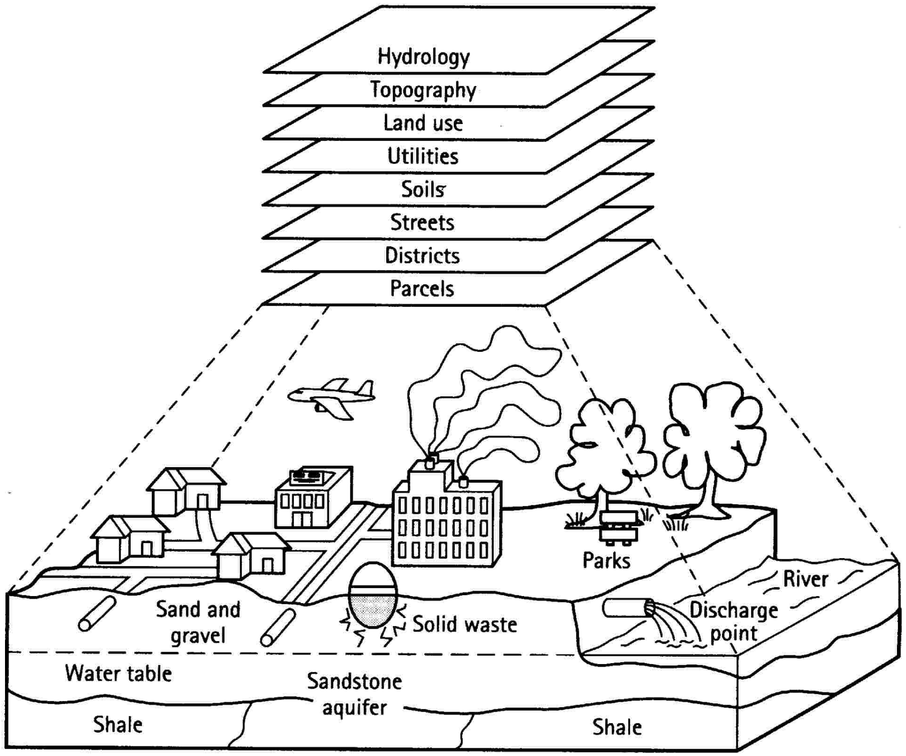
Dimensions of geoinformation

- 3D space – coordinates, address
- Time – clock time and date, change, cause, effect
- Theme – spatial related topics
- Scale – level of detail, resolution, map scale
- Quality – discrepancy between model and reality

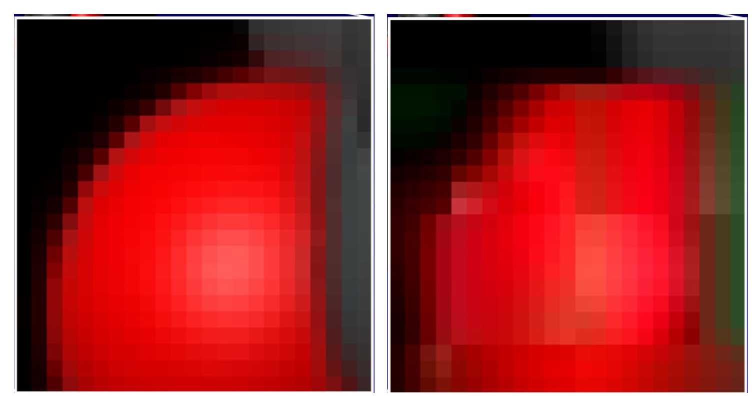
scales



themes

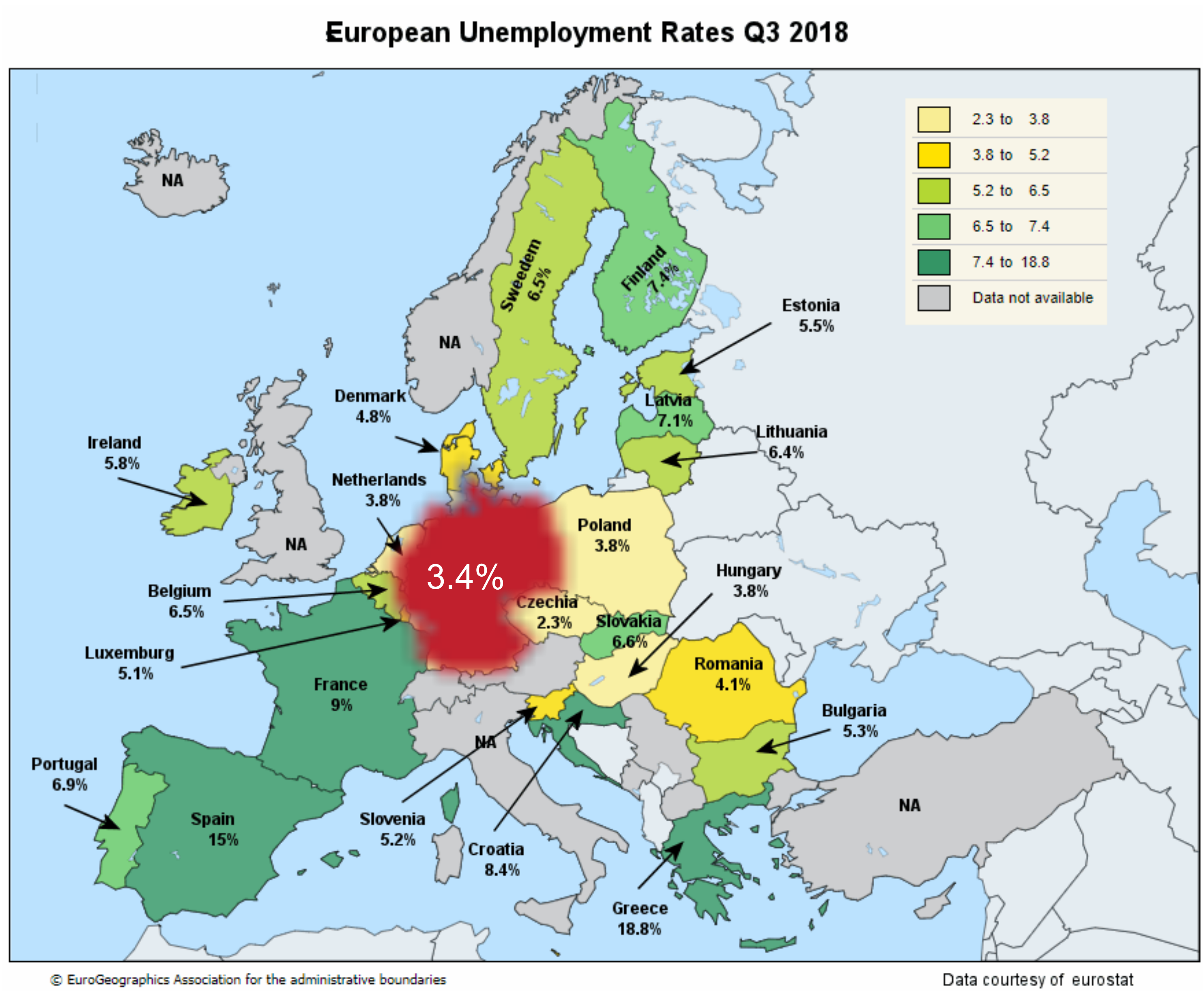


resolutions



What makes geoinformation special?

- Spatial and temporal dependence – geographic objects are interrelated.
- Spatial heterogeneity - Conditions at different places are different.
- User interest is place-sensitive – Information of a place is of greater importance to users living in that place than it is to users elsewhere.


European Unemployment Rates Q3 2018
2.3 to 3.8
3.8 to 5.2
5.2 to 6.5
6.5 to 7.4
7.4 to 18.8
Data not available
NA
Sweedem 6.5%
Finland 7.4%
Estonia 5.5%
NA
Denmark 4.8%
Latvia 7.1%
Lithuania 6.4%
Ireland 5.8%
Netherlands 3.8%
Poland 3.8%
NA
3.4%
Hungary 3.8%
Belgium 6.5%
Czechia 2.3%
Slovakia 6.6%
Luxemburg 5.1%
Romania 4.1%
France 9%
Bulgaria 5.3%
NA
Portugal 6.9%
NA
Spain 15%
Slovenia 5.2%
Croatia 8.4%
Greece 18.8%
© EuroGeographics Association for the administrative boundaries
Data courtesy of eurostat

1. How do you understand geoinformatics and its development?
2. Explain the essential components and typical workflow of a GIS.
3. What is your opinion about big geospatial data and its development?
4. What is your understanding of value cycle from data to intelligence?
5. What makes geoinformation special?
6. Use examples to explain the terms - entities, fields and phenomena.